ASSIGNMENT REPORT – 8

LDA, KNN, SVM, DTW AND PCA-BASED COMPARISON

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For The Subject

Pattern Recognition and Machine Learning (26AI5707)

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INTRODUCTION

The objective of Assignment 8 is to work with different datasets and classifiers and draw inferences with respect to both the dataset and the classifier used. The supplied assignment specifies four datasets—Synthetic, MNIST, TI digits (sequential), and Image dataset—and asks for GMM, LDA, K-NN and SVM classification, with DTW specifically for the digit sequence data. The same experiments must then be repeated after PCA projection onto the top K dimensions.

The main question is not simply which classifier has the highest accuracy, but why the ranking changes across datasets. Fixed-length image vectors, compact engineered features, synthetic distributions and time-varying speech sequences contain different types of structure. A model that is well matched to one representation can therefore be weaker on another.

1. DATASETS

Dataset	Train	Validation	Test	Representation
Synthetic	14,000	3,000	3,000	16-D fixed-length
MNIST	14,911	3,698	3,102	784-D flattened 28×28 grayscale images
TIDIGITS	945	201	204	39-D frame sequences; 312-D pooled representation for fixed classifiers
Image Features	2,000	500	500	36-D feature vectors

---Synthetic data contains two classes and 20,000 samples in total. The split used in the notebook is stratified into 14,000 training, 3,000 validation and 3,000 test samples, with 7,000/1,500/1,500 samples per class respectively.

--MNIST contains the three assigned digits 1, 6 and 9. Images are converted to grayscale, resized to 28×28 and flattened to 784 dimensions. The test set contains 1,135 digit-1 images, 958 digit-6 images and 1,009 digit-9 images.

---TIDIGITS is sequential. The training, validation and test splits contain 315/67/68 samples for each of the three labels 1, 6 and 7. The sequence representation has 39 features per frame. For GMM, LDA, KNN and SVM, a pooled fixed-length representation is used; DTW operates directly on the sequences.

--The Image Feature dataset contains five balanced classes. Each split has 36-dimensional feature vectors, with 400 training, 100 validation and 100 test samples per class

2. METHODOLOGY

2.1 Preprocessing

For the fixed-length classifiers, the notebook standardizes the training and evaluation matrices using statistics computed from the training data. This avoids letting test-set statistics influence the transformation. PCA is also fitted only on the training feature matrix and then applied to the test matrix.

2.2 Gaussian Mixture Model (GMM)

A class-conditional GMM models each class by a mixture of Gaussian components. In the implementation, two components per class are fitted using EM for a maximum of 20 iterations. The E-step computes component responsibilities, while the M-step updates mixture parameters. Classification is performed by comparing the class-conditional likelihoods.

2.3 Linear Discriminant Analysis (LDA)

LDA seeks a discriminative linear projection by using within-class and between-class scatter information. The implementation estimates the shared within-class structure and uses regularization to maintain numerical stability. A test vector is assigned according to the resulting discriminant scores.

2.4 K-Nearest Neighbours (KNN)

KNN classifies a sample according to the labels of its nearest training samples. The experiment uses k=5 and Euclidean distance. Distance computation is performed in chunks so that large datasets such as MNIST can be evaluated without constructing an unnecessarily large full distance matrix.

2.5 Support Vector Machine (SVM)

A linear one-vs-rest SVM is trained using hinge-loss gradient descent. The implementation uses learning rate 0.01, 25 epochs, C=0.05 and minibatches of size 256. The predicted class is the one with the highest linear decision score.

2.6 Dynamic Time Warping (DTW)

DTW is applied only to TIDIGITS because the data are sequential. Each utterance is resampled to 40 frames. The local cost is squared Euclidean distance and a Sakoe–Chiba window of 8 constrains the warping path. The classifier stores 25 prototypes per class and assigns each query to the class of the closest prototype under DTW distance.

2.7 PCA

PCA is implemented manually using the training covariance matrix and eigendecomposition. The eigenvectors are ordered by decreasing eigenvalue and the first K components are retained. The requested K values are dataset dependent: Synthetic uses K=5 and 10; MNIST uses K=5, 10, 20, 50 and 100; TIDIGITS and Image Features use the values shown in the results tables.

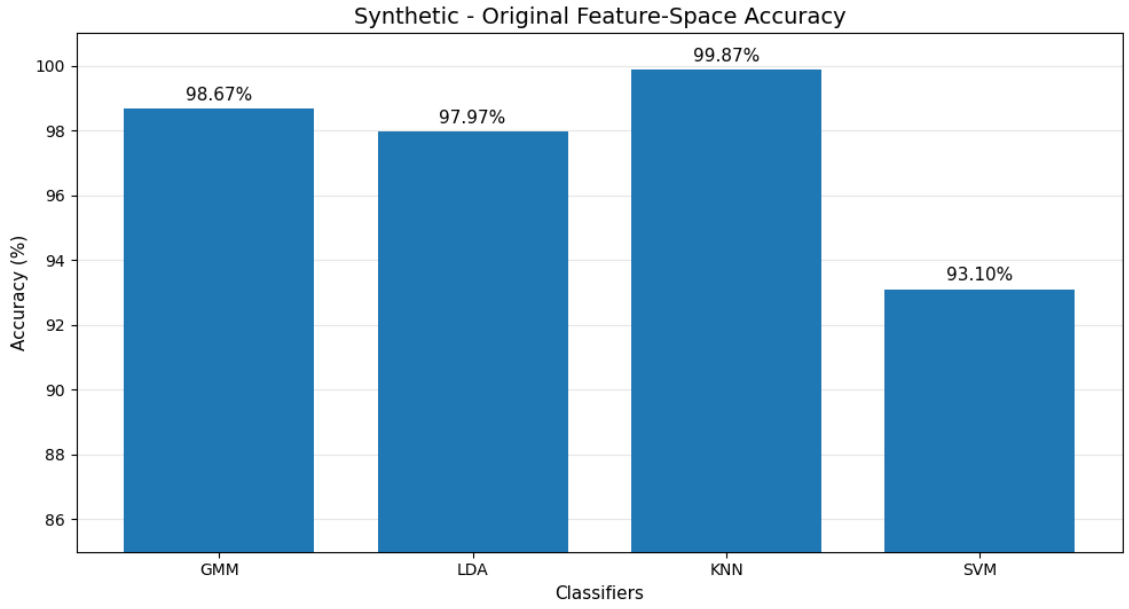
3. ORIGINAL FEATURE-SPACE RESULTS

Dataset	GMM	LDA	KNN	SVM	DTW
Synthetic	98.67%	97.97%	99.87%	93.10%	—
MNIST	96.68%	98.61%	99.03%	97.74%	—
TIDIGITS	97.55%	100.00%	99.02%	91.18%	99.02%
Image Features	59.80%	62.60%	55.40%	49.80%	—

The original-space comparison immediately shows that no single classifier dominates all datasets. KNN is best on Synthetic at 99.87% and MNIST at 99.03%. LDA is best on TIDIGITS at 100.00%. On Image Features, LDA gives the highest result at 62.60%, although the overall performance is much lower than for the other datasets.

2.8 Synthetic Dataset

Figure – Synthetic original-space classifier comparison

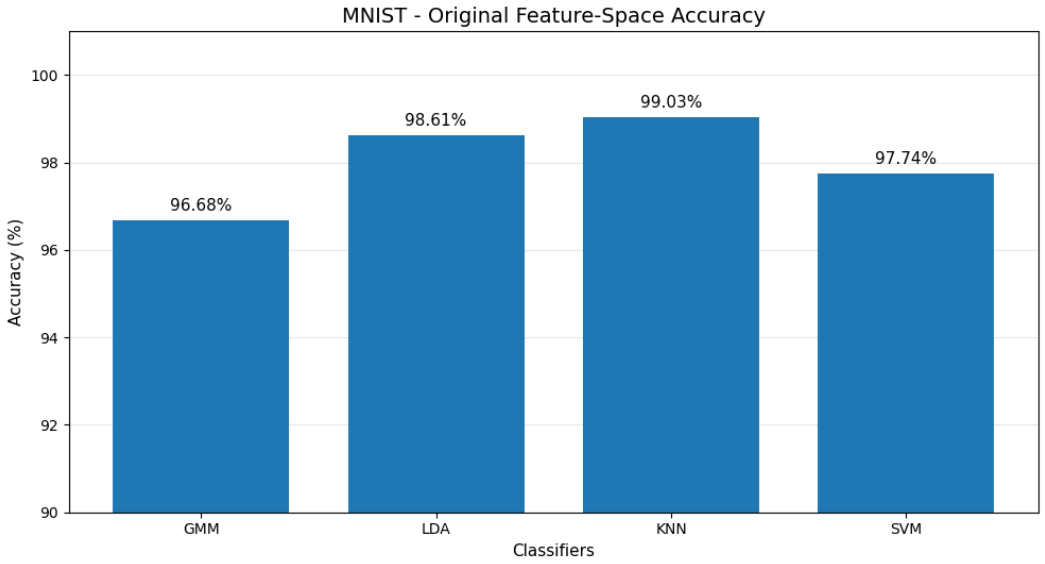


KNN achieves the highest accuracy of 99.87%, followed by GMM at 98.67% and LDA at 97.97%. SVM is lower at 93.10%. The very high KNN result suggests that local neighbourhood structure is highly informative in this synthetic feature space. The relatively lower SVM result indicates that a single linear decision function does not capture the class geometry as effectively as the neighbourhood-based model.

2.9 MNIST Dataset

KNN reaches 99.03% and is the strongest classifier in the original 784-dimensional space. LDA also performs strongly at 98.61%, while SVM obtains 97.74%. GMM got 96.68%.

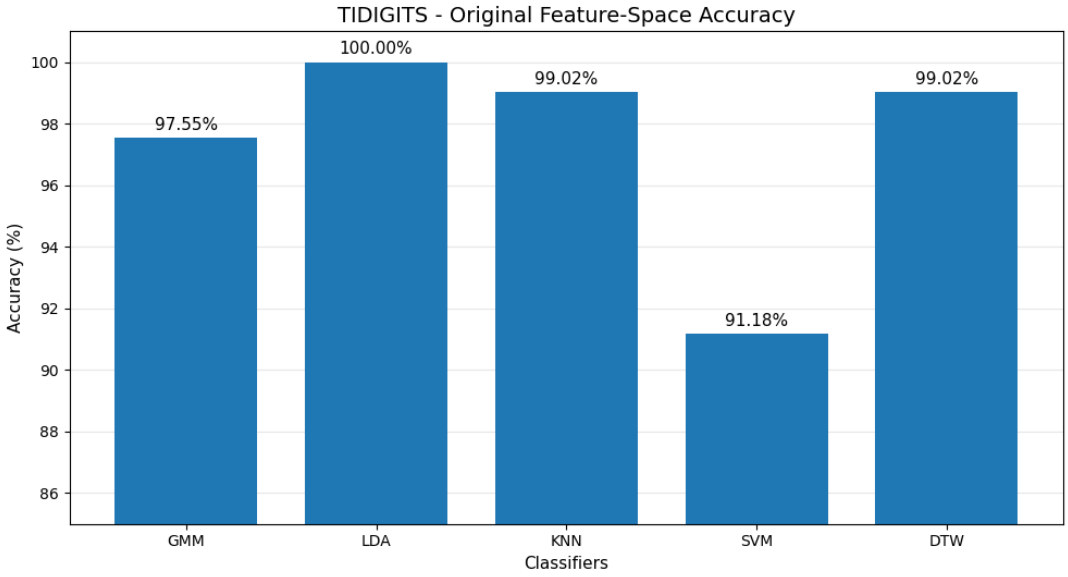
Figure – MNIST original-space classifier comparison



2.10 TIDIGITS Dataset

LDA reaches 100.00% on the fixed-length TIDIGITS representation. KNN follows at 99.02%, GMM at 97.55% and SVM at 91.18%. DTW gives 99.02%, which is only slightly below LDA. The strong performance of all but SVM indicates that the three digit classes are highly separable in the available acoustic feature representation.

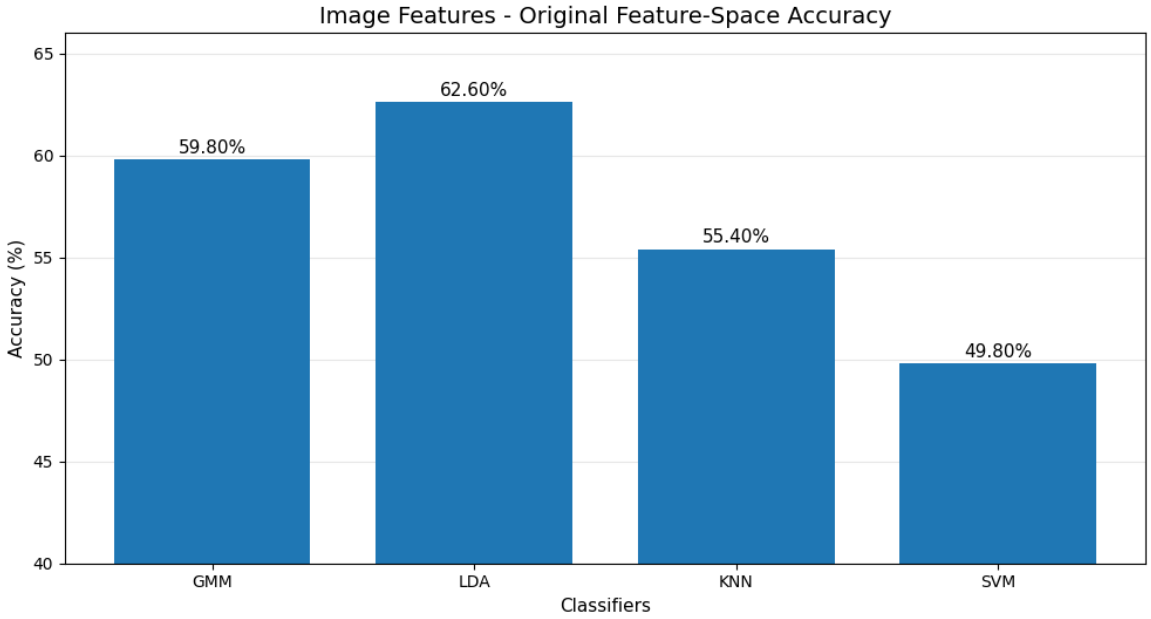
Figure – TIDIGITS original-space classifier comparison



2.11 Image Feature Dataset

The Image Feature dataset is the most difficult of the four. LDA achieves the highest accuracy at 62.60%, followed by GMM at 59.80%, KNN at 55.40% and SVM at 49.80%. The relatively small differences between several methods, together with the much lower absolute accuracy, indicate that the feature representation itself may not provide sufficiently discriminative information for these classes

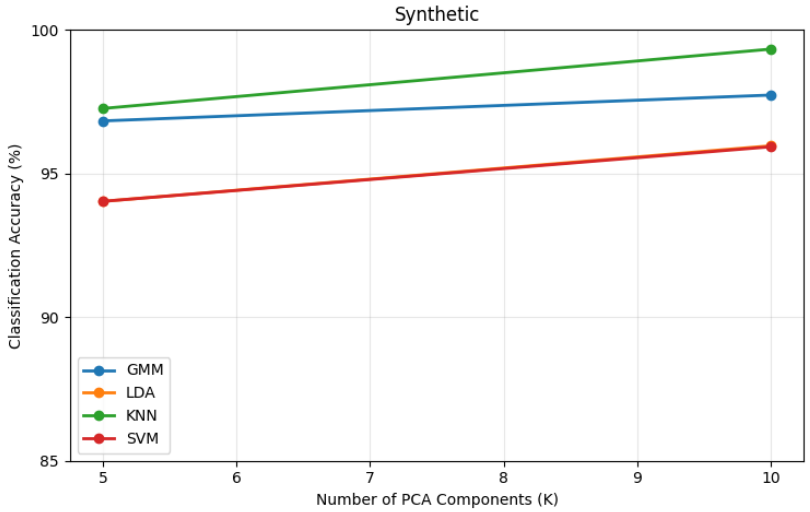
Figure – Image Feature Dataset original-space classifier comparison



1. PCA RESULTS:

1.1 Synthetic Dataset

Figure – Synthetic accuracy across PCA dimensions

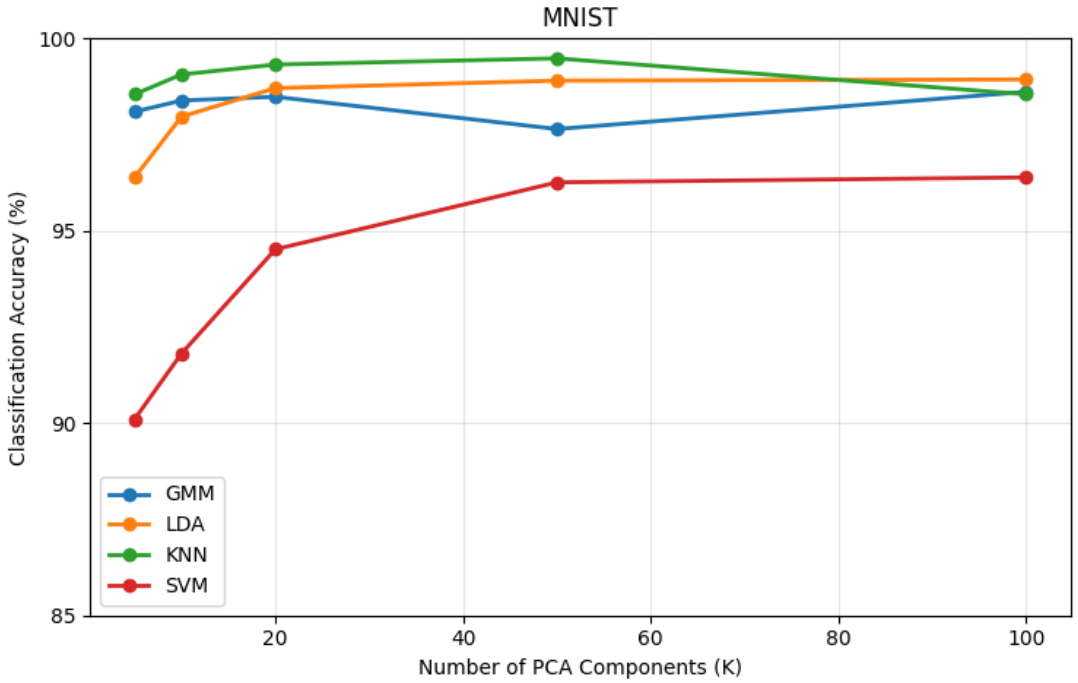


K	GMM	LDA	KNN	SVM
5	96.83%	94.03%	97.27%	94.03%
10	97.73%	95.97%	99.33%	95.93%

At K=10, KNN improves to 99.33%, which is close to the original 99.87%. GMM and both linear methods also improve compared with K=5. This shows that a compact projection can preserve most of the useful class structure while removing some dimensions that are less useful to the classifiers.

1.2 MNIST Dataset

Figure – MNIST accuracy across PCA dimensions

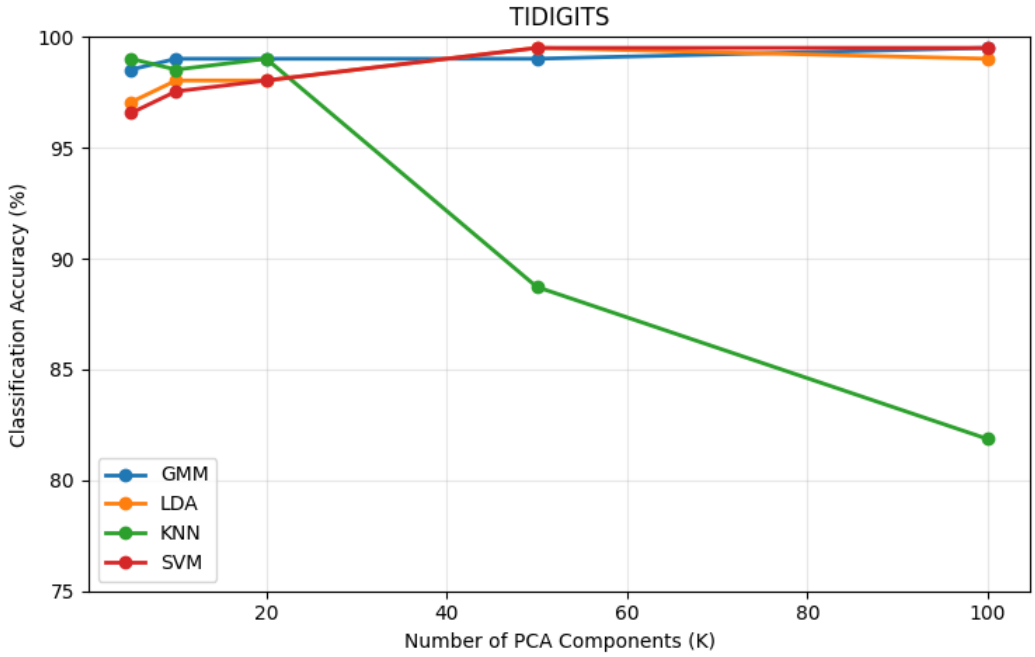


K	GMM	LDA	KNN	SVM
5	98.10%	96.39%	98.55%	90.10%
10	98.39%	97.97%	99.07%	91.81%
20	98.48%	98.71%	99.32%	94.52%
50	97.65%	98.90%	99.48%	96.26%
100	98.61%	98.94%	98.55%	96.39%

KNN is consistently the strongest MNIST model. Its best result is 99.48% at K=50, slightly higher than the original-space 99.03%. GMM remains close to 98–99% across the tested dimensions. LDA improves steadily from 96.39% at K=5 to 98.94% at K=100. SVM improves from 90.10% at k=5 to 96.39% at k=100

1.3 TIDIGITS Dataset

Figure – TIDIGITS accuracy across PCA dimensions

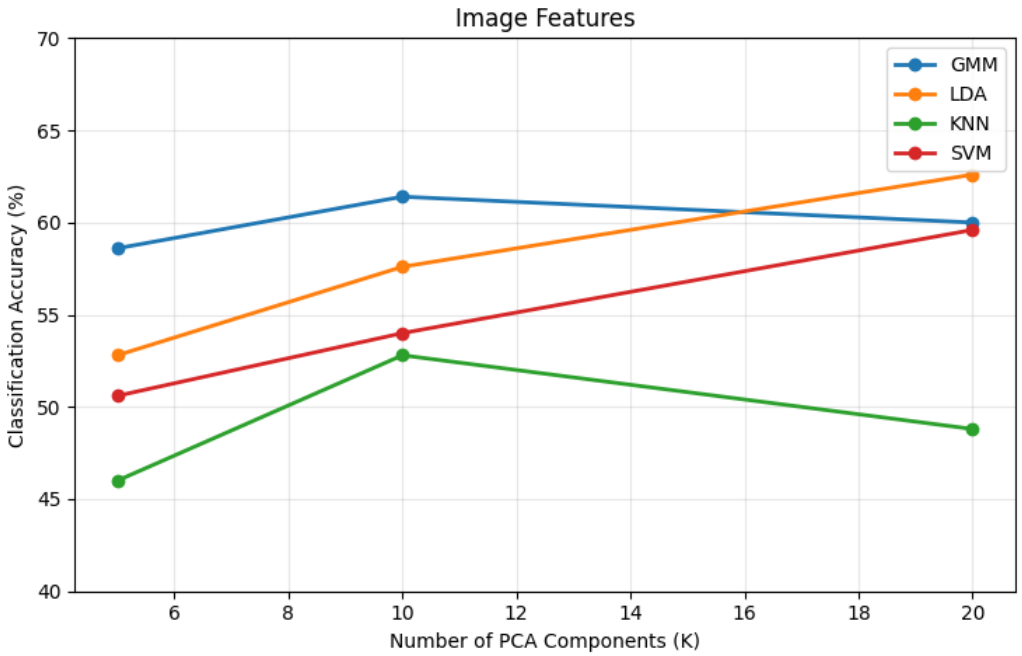


K	GMM	LDA	KNN	SVM
5	98.53%	97.06%	99.02%	96.57%
10	99.02%	98.04%	98.53%	97.55%
20	99.02%	98.04%	99.02%	98.04%
50	99.02%	99.51%	88.73%	99.51%
100	99.51%	99.02%	81.86%	99.51%

TIDIGITS shows a strong dependence on K for KNN. At K=5, KNN gives 99.02%, but it falls to 81.86% at K=100. In contrast, LDA and SVM reach 99.51% at K=50, while GMM reaches 99.51% at K=100. Thus, retaining more dimensions is beneficial for some classifiers but harmful for KNN. This is a clear example of the interaction between representation dimensionality and classifier behaviour.

1.4 Image Feature Dataset

Figure – Image Features accuracy across PCA dimensions



K	GMM	LDA	KNN	SVM
5	58.60%	52.80%	46.00%	50.60%
10	61.40%	57.60%	52.80%	54.00%
20	60.00%	62.60%	48.80%	59.60%

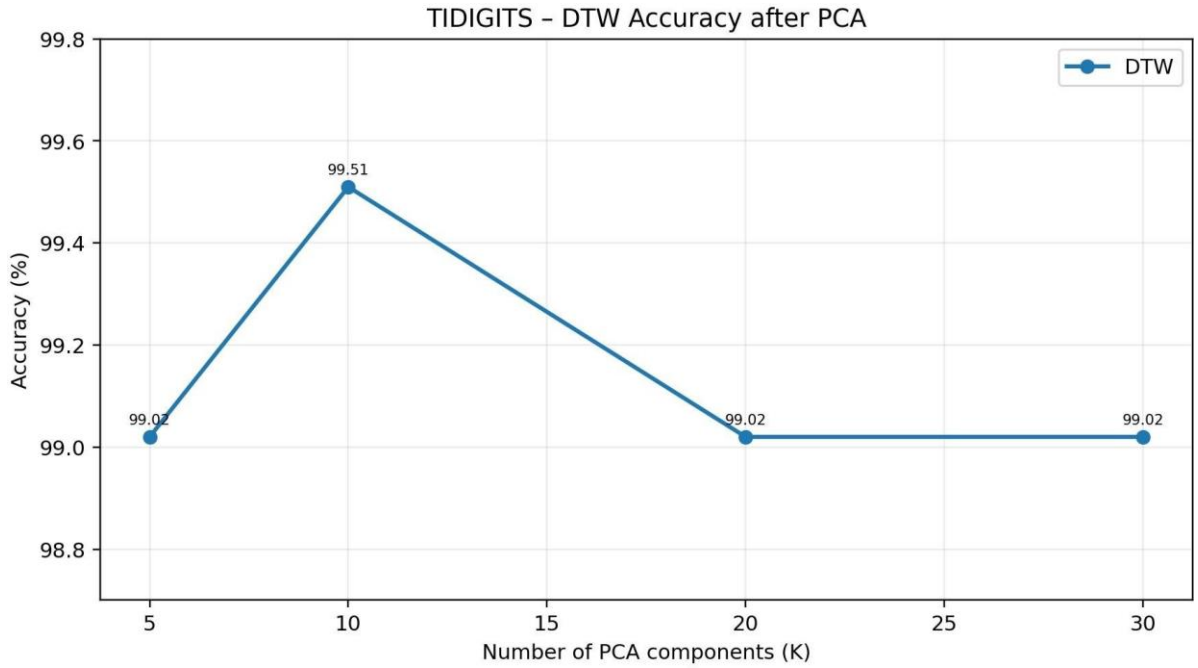
PCA gives mixed results for Image Features. LDA reaches the original-space value of 62.60% at K=20, while GMM reaches 61.40% at K=10. KNN improves from 46.00% at K=5 to 52.80% at K=10 but falls to 48.80% at K=20. SVM improves gradually to 59.60% at K=20. These results reinforce that PCA is not automatically beneficial; the retained subspace must preserve information useful to the particular classifier.

2. DTW WITH PCA ON TIDIGITS

PCA components	DTW accuracy
5	99.02%
10	99.51%
20	99.02%
30	99.02%

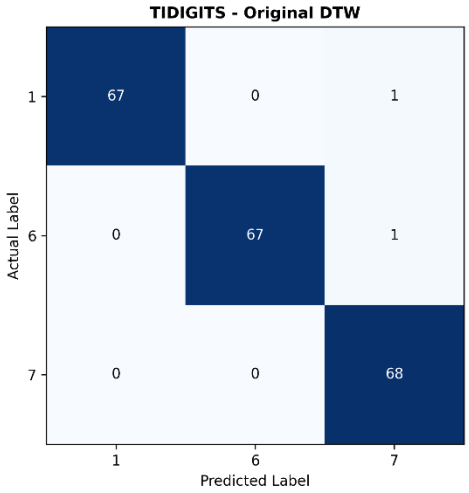
For DTW, PCA is applied at the frame level. PCA is fitted using all training frames, each utterance is projected frame-by-frame, and DTW is then performed on the projected sequences. The best result is 99.51% at K=10. This improves on the 99.02% obtained using the original frame features and demonstrates that a compact frame representation can preserve the temporal structure needed for DTW while reducing irrelevant variation.

Figure – TIDIGITS DTW accuracy after PCA



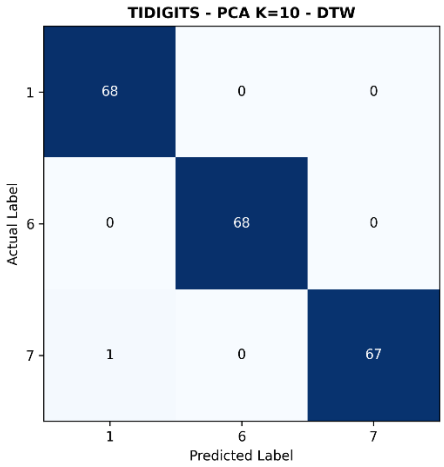
3. CONFUSION-MATRIX ANALYSIS

3.1 TIDIGITS – DTW (Original)

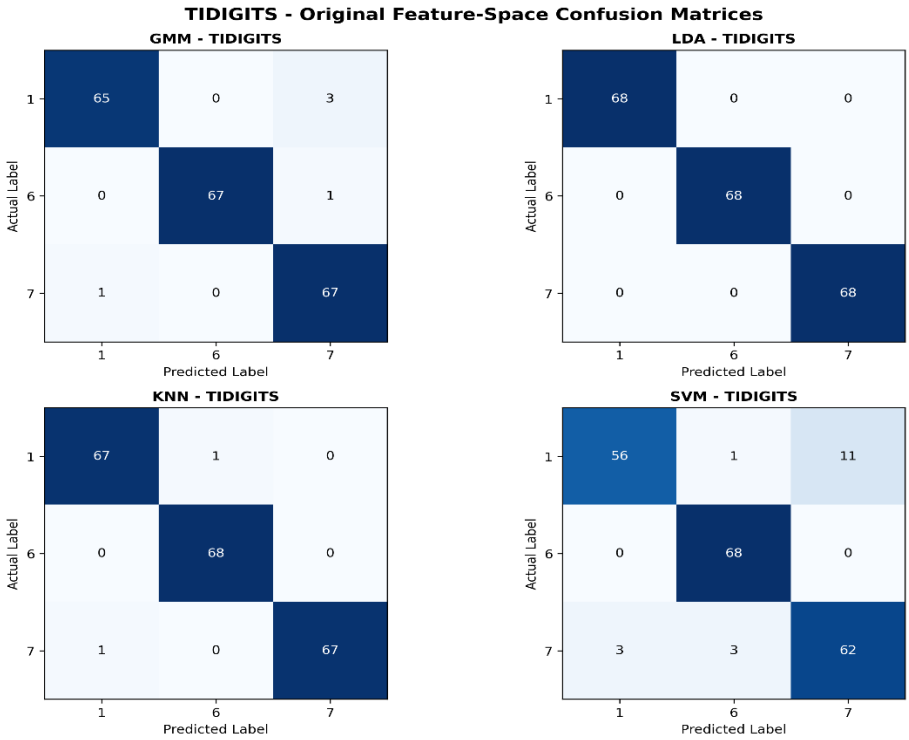


The diagonal entries represent correctly classified samples, while off-diagonal entries show the error patterns. For TIDIGITS, the DTW matrices are almost completely diagonal, which agrees with the 99%+ accuracies. The Image Feature matrix shows more substantial confusion among classes, especially involving class 2 and class 4. For MNIST KNN at K=50, the matrix is strongly diagonal, explaining its 99.48% accuracy.

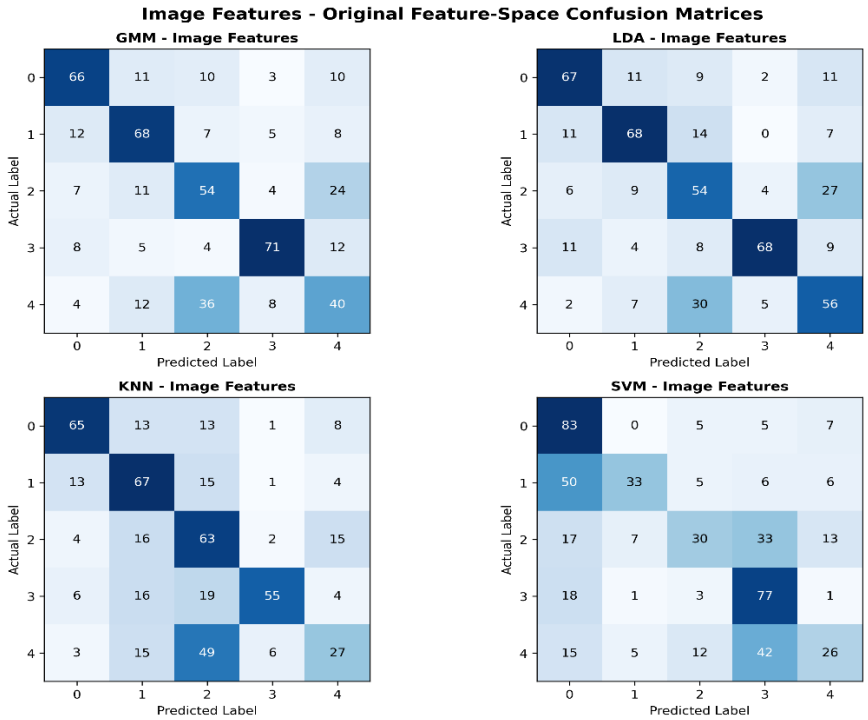
3.2 TIDIGITS – DTW (PCA K=10)



The diagonal entries represent correctly classified samples, while off-diagonal entries show the error patterns. For TIDIGITS, the DTW matrices are almost completely diagonal, which agrees with the 99%+ accuracies. The Image Feature matrix shows more substantial confusion among classes, especially involving class 2 and class 4. For MNIST KNN at K=50, the matrix is strongly diagonal, explaining its 99.48% accuracy.

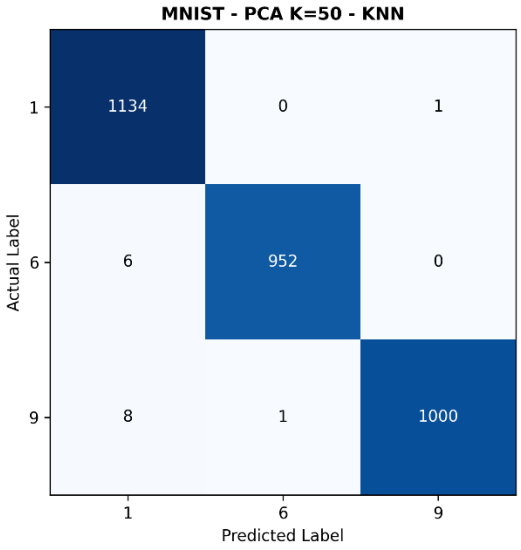


3.3 Image Features – LDA (Original)

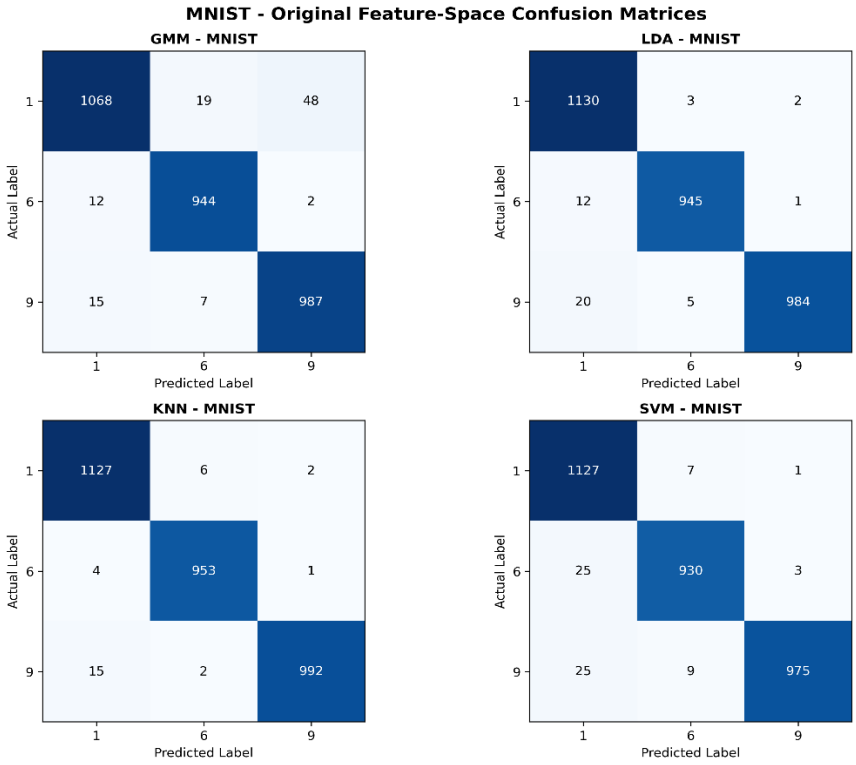


The diagonal entries represent correctly classified samples, while off-diagonal entries show the error patterns. For TIDIGITS, the DTW matrices are almost completely diagonal, which agrees with the 99%+ accuracies. The Image Feature matrix shows more substantial confusion among classes, especially involving class 2 and class 4. For MNIST KNN at K=50, the matrix is strongly diagonal, explaining its 99.48% accuracy.

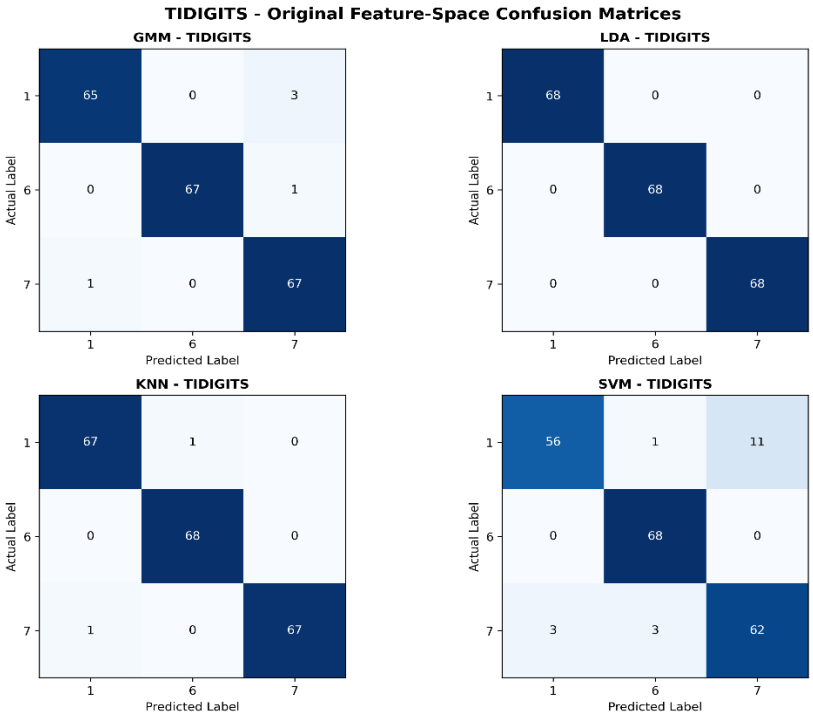
3.4 MNIST – KNN (PCA K=50)



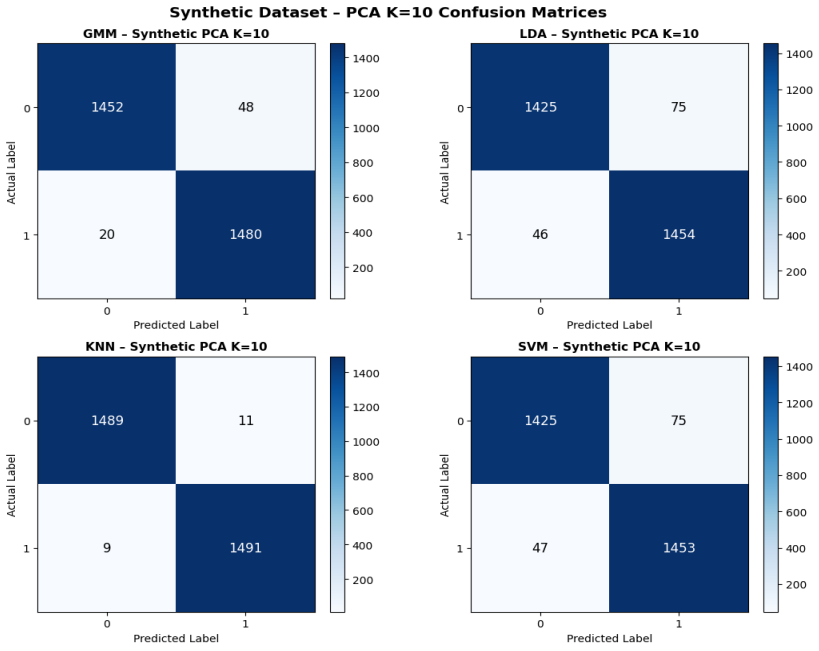
The diagonal entries represent correctly classified samples, while off-diagonal entries show the error patterns. For TIDIGITS, the DTW matrices are almost completely diagonal, which agrees with the 99%+ accuracies. The Image Feature matrix shows more substantial confusion among classes, especially involving class 2 and class 4. For MNIST KNN at K=50, the matrix is strongly diagonal, explaining its 99.48% accuracy.



3.5 Synthetic



The diagonal entries represent correctly classified samples, while off-diagonal entries indicate misclassification patterns. For the Synthetic dataset, all four classifiers achieve high classification accuracy, with most samples correctly classified across both classes. KNN produces the strongest performance, followed by GMM and LDA, while SVM shows comparatively more misclassifications. These results indicate that the synthetic data classes are highly separable, and neighbourhood-based classification performs particularly well in this feature space.



4. COMPARATIVE DISCUSSION

The experiments demonstrate that classifier performance is dataset dependent. KNN is particularly effective when nearby samples belonging to the same class form coherent local regions. This is visible on the Synthetic and MNIST datasets. GMM is also strong on these datasets, but its probabilistic assumptions do not always match the discriminative structure as well as KNN.

LDA performs exceptionally well on TIDIGITS, reaching 100% before PCA and 99.51% at its best PCA setting. However, the same method gives only 36.59% on MNIST. This contrast is important: the effectiveness of a discriminant model depends on whether its assumptions about class structure are appropriate for the feature representation.

SVM provides competitive MNIST performance and improves substantially as more PCA components are retained. Its 91.18% original TIDIGITS result is weaker than the other methods, but it reaches 99.51% at K=50 and K=100 after PCA. On Image Features it remains below LDA and GMM, although its performance improves from 49.80% to 59.60%.

DTW is especially appropriate for sequential data because it explicitly handles temporal misalignment. Its 99.02% original accuracy and 99.51% at K=10 show that temporal matching is highly effective for the TIDIGITS task. The result is also comparable to the strongest fixed-length classifiers.

PCA should therefore be viewed as a representation-selection tool rather than a guaranteed accuracy-improvement step. It can remove redundancy and noise, but if K is too small, discriminative information can be lost. Conversely, retaining many dimensions can preserve useful information for one classifier while causing a different classifier to suffer, as seen clearly for TIDIGITS KNN.

5. BEST CONFIGURATIONS

Dataset	**Best method/configuration**	**Accuracy**
Synthetic	KNN, original space	99.87%
MNIST	KNN, PCA K=50	99.48%
TIDIGITS	LDA, original space	100.00%
TIDIGITS	DTW, PCA K=10	99.51%
Image Features	LDA, original space / PCA K=20	62.60%

6. CONCLUSION

Assignment 8 provided a comparative study of GMM, LDA, KNN, SVM and DTW across fixed-length and sequential datasets, followed by PCA-based dimensionality reduction. The results confirm that there is no universally best classifier. KNN provides the strongest performance on the Synthetic and MNIST datasets, LDA is particularly effective on TIDIGITS, and DTW is highly competitive for sequential speech features. The Image Feature dataset remains the most challenging, with all classifiers below 63%.

The PCA experiments further show that dimensionality reduction must be selected with the classifier and dataset in mind. The best observed results were 99.33% for Synthetic KNN at K=10, 99.48% for MNIST KNN at K=50, 99.51% for TIDIGITS GMM/LDA/SVM at suitable K values, and 99.51% for TIDIGITS DTW at K=10. Overall, the experiments demonstrate the importance of matching the model and feature representation to the underlying structure of the data.